

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO1: Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1,BL2
Assessment Tool Weightage: 30%

QUESTIONS: 5

Total Marks:25

Concept Mapping
Annexure A

Criteria	Concept Identification (2.5)	Linkages (2.5)	Organization (2)	Integration of Literature (2)	Clarity (1)	Total(10)
Weightage	25%	25%	20%	20%	10%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

*I J.Raju(192325117) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

J.Raju

RUBRICS FOR CALCULATION:

Concept Mapping					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

Annexure A

Concept Map 1: Decision Support Systems (DSS)

Task Description:

Construct a concept map that explains how a Decision Support System (DSS) supports organizational decision-making.

What your map must show:

1. Core components: Data, Model, User Interface
 2. Types of DSS
 3. Flow: Data → Analysis → Decision
 4. Role in business decision-making
 5. Real-world example linkage
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Concept Map 2: Operational Systems vs DSS

Task Description:

Develop a comparative concept map that clearly distinguishes Operational Systems (OLTP) and Decision Support Systems (OLAP).

What your map must show:

1. Definitions of OLTP and DSS
2. Differences (data type, purpose, users, processing)
3. Similarities and interactions
4. Real-time vs historical data
5. Example systems (banking vs analytics)

Concept Map 3: Architecture of Data Warehouse

Task Description:

Create a hierarchical concept map illustrating the architecture of a Data Warehouse.

What your map must show:

1. Data Sources → ETL → Data Warehouse → Data Marts

2. Metadata layer
3. Front-end tools (reporting, OLAP)
4. Data flow direction
5. Role of each layer

Concept Map 4: OLAP, Data Cubes & Dimensional Modeling

Task Description:

Design a concept map that connects OLAP operations, Data Cubes, and Dimensional Modeling techniques.

🌀 What your map must show:

1. Data cube structure (dimensions, measures)
2. OLAP operations (slice, dice, roll-up, drill-down)
3. Star schema and snowflake schema
4. Fact and dimension tables
5. Analytical usage

Concept Map 5: Query Processing & BI Tools (KNIME)

Task Description:

Construct a concept map explaining how query processing integrates with BI tools, focusing on KNIME for reporting and OLAP applications.

What your map must show:

1. Query processing stages (Parsing, Optimization, Execution)
2. Role of BI tools
3. KNIME workflow structure
4. Data → Query → Visualization pipeline
5. OLAP application integratio

INTRODUCTION TO DATA WAREHOUSING

The Data Warehousing Ecosystem: From Raw
Data to Strategic Decisions



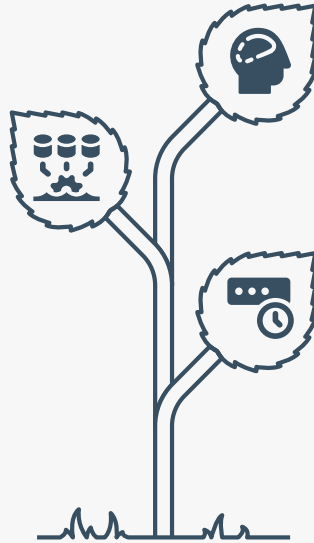
Data Warehousing Ecosystem

The ecosystem spans from raw data to strategic decisions.

Decision Support System (DSS)

Lack of Business Insights

Patterns and performance metrics are not identified



Reliance on Intuition

Data-driven insights are not prioritized over gut feelings

Limited Historical Analysis

Past performance is not leveraged for future predictions

Purpose: Enabling Strategic Decision Making through Historical Analysis

- Strategic Decision Making: Long-term planning and goal setting.
- Business Analysis: Identifying patterns and performance metrics.
- Historical Data: Leveraging past performance to predict future outcomes.
- Better Decision Making: Data-driven insights over intuition.

Operational (OLTP) vs. Decision Support (OLAP)

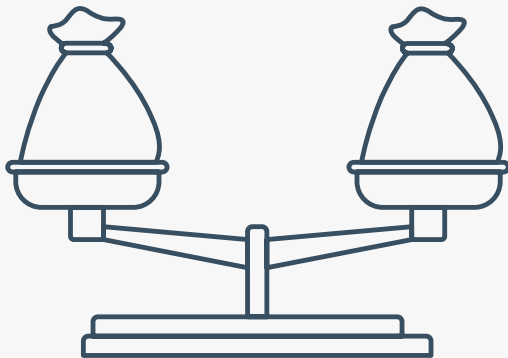
Operational System (OLTP)

Daily transactions

Focuses on daily transactions and current data.

Fast processing

Optimized for fast processing and high-frequency updates.



Decision Support System (OLAP)

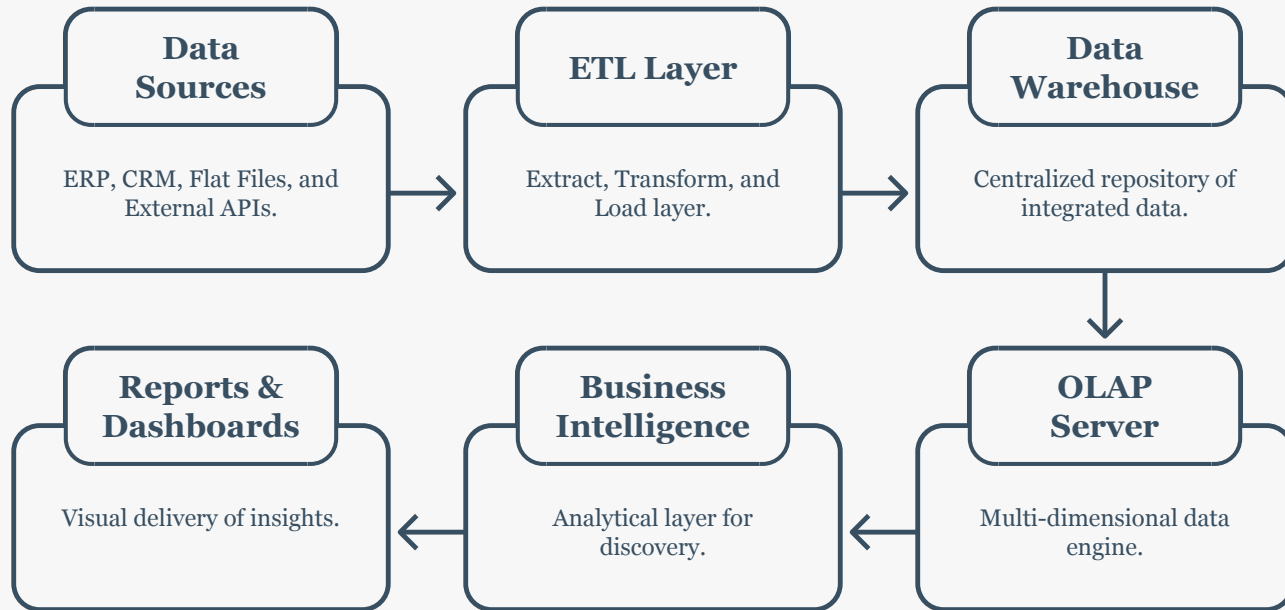
Historical data

Focuses on historical data and business intelligence.

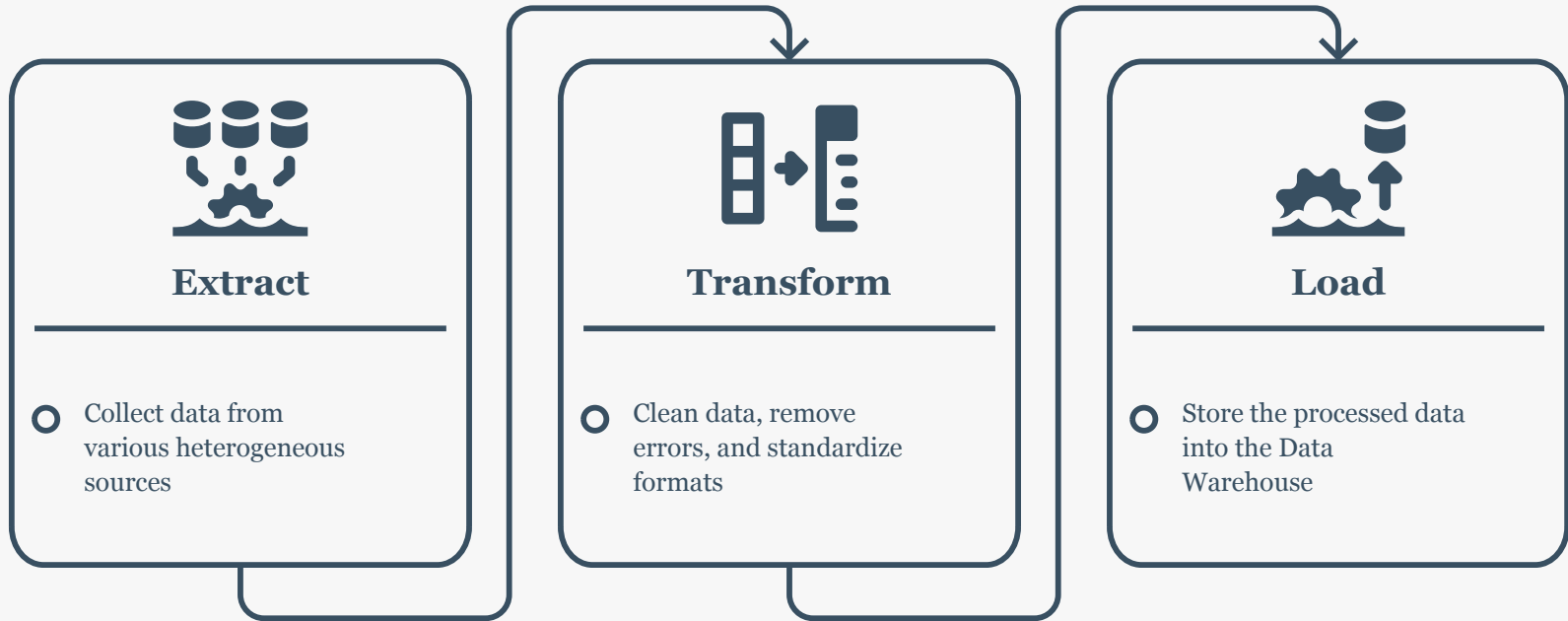
Data analysis

Optimized for data analysis, reporting, and trend analysis.

Data Warehouse Architecture: End-to-End Flow



The ETL Process



OLAP: Online Analytical Processing

Enabling Multi-dimensional Analysis and Fast Querying

- Decision Support: Providing the evidence for business choices.
- Business Intelligence: The core engine of modern BI.



OLAP

Supports multi-dimensional analysis and fast querying



OLTP

Focuses on transactional processing and real-time updates

The Data Cube



Dimensions

Dimensions: Time, Product, and Location.



Slice & Dice

Selecting specific sub-sets of data.



Drill Down & Roll Up

Changing the level of detail/granularity.



Pivot

Re-orienting the view of the cube.

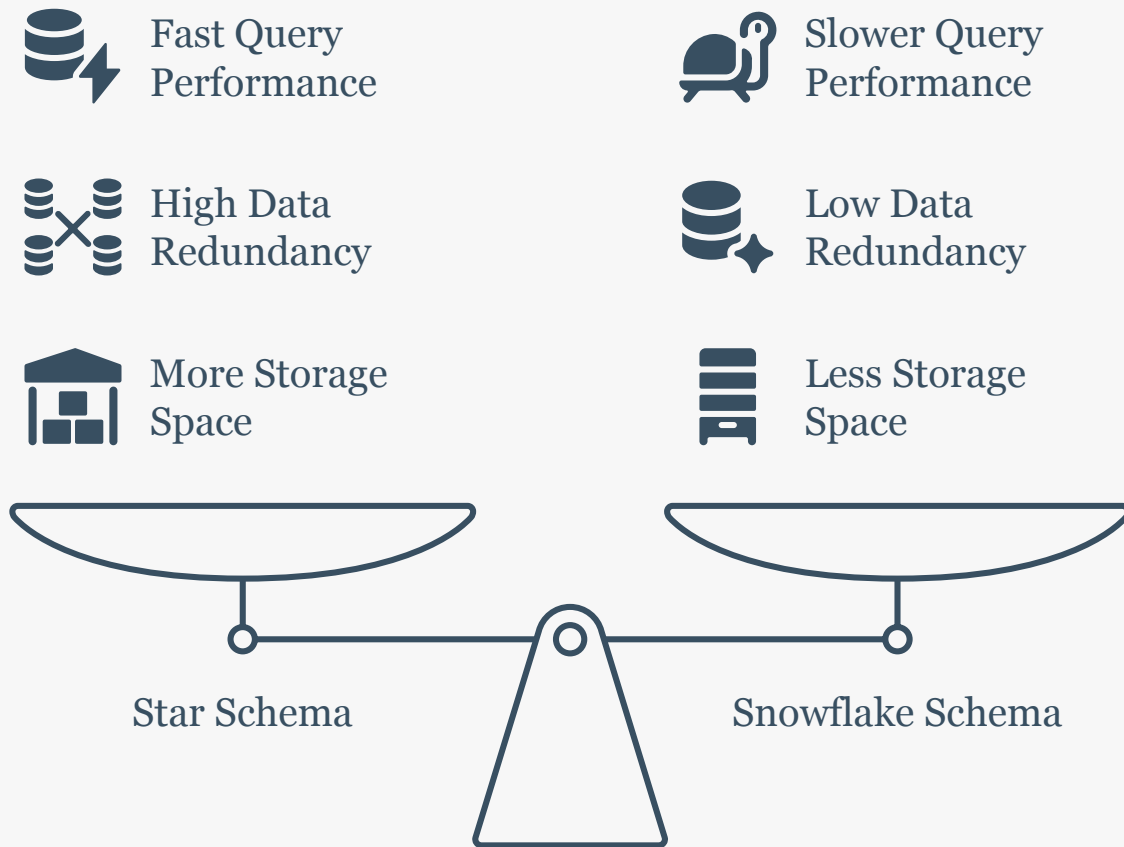
3-Dimensional Analysis and
Core Operations

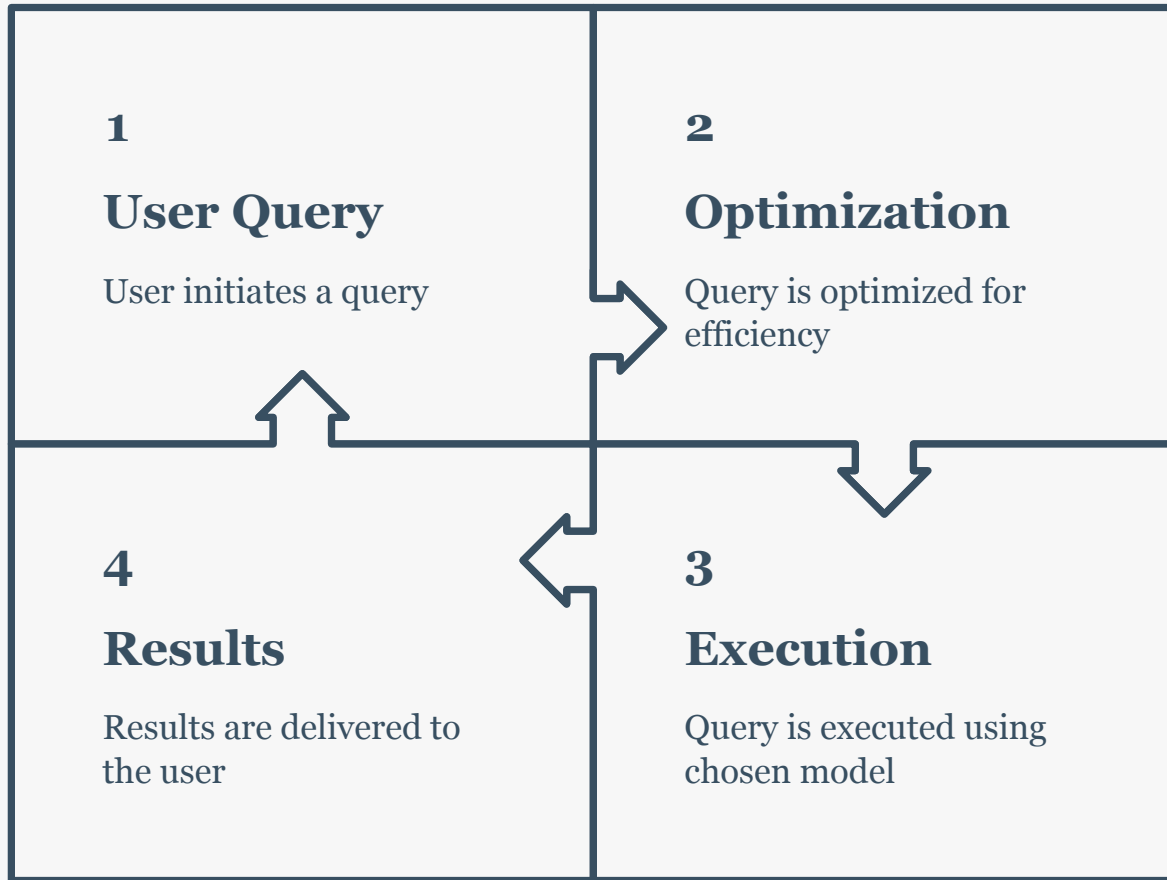
Dimensional Data Modeling

Star Schema vs. Snowflake Schema

Star Schema: A central Fact Table connected to Dimension Tables.

Snowflake Schema: Normalized Dimension Tables.





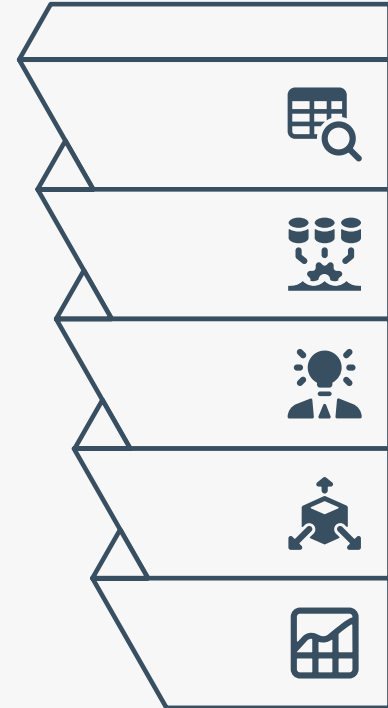
Query Processing & Computation

Computation Models: MOLAP (Multidimensional OLAP), ROLAP (Relational OLAP), and HOLAP (Hybrid OLAP).

Business Intelligence Tools & Workflows



Modern Software and the
KNIME Analytical Process



Turning Data into Smart Business Decisions



Interactive Dashboards

Real-time monitoring of business health.



KPIs

Tracking Key Performance Indicators.



Data Visualization

Making complex data understandable.



Predictive Analytics

Forecasting future trends.



Smart Business Decisions

The ultimate goal of the ecosystem.